



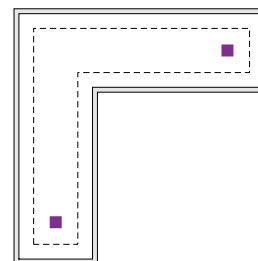
The indoor environment and sensor system

- The focus of the rest of this week is to show how a particle filter can be used to solve an indoor-navigation problem.
 - For outdoor navigation using GPS, an EKF is said to work very well;
 - For GPS-deprived indoor navigation, the added complexity of the particle filter can be justified by the greater challenge imposed by fusing measurements from diverse sensors.
- I will first describe the map of the indoor environment that I will use in the example, then the sensor platform I will assume.
- In the next lesson, I will set up the pdfs needed by the particle filter.
- Finally, two lessons from now, I will show how to implement the navigation application in Octave code and present some results.



A map of the indoor space

- Every real problem will be different; since the focus of this course is on particle filtering, the emphasis of this example is on how to solve one simple navigation problem with the filter rather than on how to solve every navigation problem.
- I have chosen to implement a room that has maximum dimensions of $W = 10$ m in both dimensions.
- A square cutout of width $F \times W = 7$ m is made from the bottom right; the remaining room is “L” shaped with 3 m width.
- Radio beacons—denoted by purple squares—are placed at user-defined points. In this figure, at (1.5,1.5) and at (7.5,7.5) if the bottom-left corner is the (0,0) point.
- The dashed line shows the path to be taken by the vehicle whose position we seek to estimate via a particle filter.



Vehicle INS and dead-reckoning sensor model

- The vehicle must be equipped with onboard inertial sensors to measure relative (changes to) position.
- I am assuming that these will measure speed s_k and direction θ_k .
 - Perhaps via wheel speed sensors and compass measurements, although the indoor environment degrades compass accuracy.
- The state equation for the vehicle is:

$$\begin{bmatrix} \xi_{k+1} \\ \eta_{k+1} \end{bmatrix} = \begin{bmatrix} \xi_k \\ \eta_k \end{bmatrix} + \Delta t \begin{bmatrix} s_k \cos \theta_k \\ s_k \sin \theta_k \end{bmatrix}$$

- The measured signals will have noise, which I have modeled as: $s_k^m = s_k + w_k^s$ and $\theta_k^m = \theta_k + w_k^\theta$.
- I will assume that: $w_k^s \sim \mathcal{N}(0, \sigma_s^2)$ and $w_k^\theta \sim \mathcal{N}(0, \sigma_\theta^2)$.



Vehicle absolute position sensing

- The vehicle must be able to measure some indicator of absolute position, which I do with aid of the radio beacons.
- I assume that the map of the room and the location of the radio beacons is known.
- As you learned in the prior lesson, there are different basic methods (trilateration and triangulation) to infer position based on a radio signal.
 - This example uses trilateration, based on the power of the radio signals received from the radio beacons—a “received signal strength indicator” (RSSI) approach.
- So, we need an equation that describes the received signal strength in terms of the distance between the vehicle and a beacon.
 - We assume that the transmit signal strength is known, so computing the received signal strength requires that we can calculate the “path loss” of the signal.
 - Much more complicated to model in an indoor versus outdoor environment, motivating use of particle filters.



Components of path loss between Tx and Rx

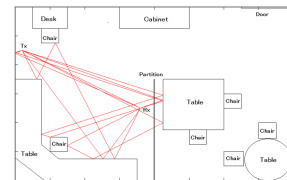
- Free-space loss is experienced in an obstacle-free line-of-sight path. It increases with square of both distance and frequency of the radio waves. In decibels (and then for a known fixed transmission frequency),

$$\text{FSPL} = 20 \log_{10}(d) + 20 \log_{10}(f) + 20 \log_{10}(4\pi/c) = (\text{loss at 1 m}) + 20 \log_{10}(d).$$
- Penetration loss experienced by the radio wave as it passes through obstacles between the antennae (e.g., walls). A constant value for any given material.
- Diffraction loss: When signals pass near sharp corners, they bend/refract, and not all of the signal goes in the straight line we might assume.
- Diffuse scattering loss: When a signal reflects off an object, some is scattered by the surface roughness; not all is reflected in the direction we might assume.
- Multipath loss: When various rays of the radio wave reflect and/or refract over the path until some arrive at the receiving antenna out of phase with the direct wave.



RSSI modeling via ray launching and tracing

- There are different approaches to computing path-loss.
- The most direct are “ray launching/tracing,” in which the signal between transmitter (Tx) and receiver (Rx) is considered to comprise a set of rays.
 - Ray launching computes from Tx to Rx; ray tracing computes from Rx to Tx.
- The losses experienced by these rays as they travel between Tx and Rx can be calculated via geometric optics: including free-space loss, ray reflection, refraction, and penetration, and fading due to multipath phase cancellation.
- Computationally demanding, but can be implemented efficiently in parallel processing and using GPUs.
- Less time-consuming and less expensive as compared to measurement campaigns.
- Accuracy depends on an accurate model of the indoor scenario, including environmental objects.





Large-scale RSSI modeling via stochastic models

- Some well-established approximate power-fade models reduce the accuracy requirements for the map of the indoor space.
 - Uncertainties are modeled using random variables.
- Models must describe large-scale fading, which refers to the attenuation of a signal due to free space and obstacles between Tx and Rx.
 - Free-space loss is computed deterministically based on distance.
 - Shadow fading (shadowing) refers to power loss due to obstructions in the path of propagation.
 - Deterministic portions might include transmission losses through walls.
 - The log-normal model is commonly used to describe random shadowing loss (in dB) as $P \sim \mathcal{N}(0, \sigma^2)$.
- Models may also describe small-scale fading, which is caused by fluctuations in signal strength over short distances and small durations in time. . .



Small-scale RSSI modeling via stochastic models

- Multipath is a “small-scale” fading phenomenon.
- Rayleigh fading describes multipath loss when there is no dominant line-of-sight path.
 - The Rayleigh distribution describes the magnitude of the sum of two uncorrelated Gaussian RVs;
 - Rayleigh fading models the received signal amplitude via the pdf

$$\mathbb{P}_R(r) = \frac{2r}{\Omega} \exp\left(-\frac{r^2}{\Omega}\right), \text{ where } \Omega = \mathbb{E}[R^2] \text{ is the average received-signal power.}$$
- Rician fading describes multipath loss when there is a line-of-sight path. If K is the ratio between power in the direct path to power in scattered paths and $I_0(\cdot)$ is the 0th-order modified Bessel function, we have (same as Rayleigh if $K = 0$):

$$\mathbb{P}_R(r) = \frac{2(K+1)r}{\Omega} \exp\left(-K - \frac{(K+1)r^2}{\Omega}\right) I_0\left(2\sqrt{\frac{K(K+1)}{\Omega}}r\right).$$



Vehicle absolute position sensing (cont.)

- For the purpose of the simple indoor-navigation example, I chose to use a stochastic model of RSSI.
- This model included free-space loss, plus deterministic shadowing loss through walls, plus stochastic shadowing loss via the log-normal model.
 - I did not include a multipath fading (e.g., Rayleigh or Rician) model, but that is something that you could add if you chose to do so.
- The overall power loss (in dB) between Tx and Rx was modeled as:

$$\text{Loss} = (\text{loss at 1 m}) + 20 \log_{10}(d) + N_w \times (\text{loss per wall}) + S,$$

where random shadowing is modeled via $S \sim \mathcal{N}(0, \sigma_{In}^2)$.



Summary

- In this lesson, you learned the details of the map of the indoor space I will use in the indoor-navigation example.
- You learned the details of the vehicle inertial sensors and position model that I use.
- In preparation to understand the absolute-position sensing method I chose, you learned about methods to compute RSSI.
 - Power losses between Tx and Rx include free-space losses, penetration losses, diffraction losses, diffuse scattering losses, and multipath losses.
 - These losses can be modeled via deterministic ray-tracing, or stochastic methods.
 - Stochastic terms include Rayleigh fading, Rician fading, and/or shadowing.
- I chose to implement a stochastic method that includes free-space loss, wall-penetration losses, and stochastic shadowing via the log-normal model.
- In the next lesson, you will learn how to leverage these models to derive the pdfs needed to implement the particle filter.



Credits

Credits for images in this lesson

- The “ray tracing” figure on slide 6 is from the open-access article: Md Humayun Kabir and Ryuji Kohno. “A hybrid TOA-fingerprinting based localization of mobile nodes using UWB signaling for non line-of-sight conditions.” *Sensors*, 12.8 (2012): 11187–11204. <https://doi.org/10.3390/s120811187>.